# Profit Analysis due to Migration Asymmetry in a Two-Patch Fishery Model

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**Abstract**

In this study, we investigate the effects of spatial heterogeneity and migration asymmetry on the economic performance of a two-patch fishery system. We develop and analyze a spatially explicit bioeconomic model that incorporates logistic growth, symmetric harvesting effort, and asymmetric dispersal between two patches. Particular attention is given to the role of temperature-dependent migration asymmetry, modeled as a linear function of environmental temperature. Using numerical simulations and equilibrium analysis, we explore how variations in fishing effort, cost, and migration influence the spatial distribution of biomass and total profit. We derive an explicit condition for the existence of a profitable equilibrium and compute the critical fishing effort ,beyond which population collapse occurs. The results demonstrate that MEY can be substantially increased by allowing moderate levels of asymmetric migration, especially when patches differ in productivity. Moreover, we show that while excessive fishing effort or dispersal can destabilize the system, a carefully optimized balance between cost, migration asymmetry, and environmental parameters yields maximal long-term economic benefit. Our findings emphasize the importance of integrating spatial and ecological heterogeneity into fishery management strategies to achieve both sustainability and profitability.

## 1 Introduction

The management of marine resources critically depends on understanding spatial heterogeneity and organism dispersal, as these factors significantly influence population dynamics and economic outcomes [1]. Classical fishery models, such as the Schaefer model [2], typically neglect spatial structure by assuming homogeneous environments and uniform harvesting strategies. However, natural ecosystems exhibit distinct spatial patchiness, with biological and economic parameters often varying substantially across different locations [3]. Furthermore, migration—often asymmetric—redistributes biomass between these patches, introducing complexities that can lead to non-intuitive outcomes in sustainability and profitability [4].

Recent research by Elbetch et al. [7] has demonstrated that while linking two spatially distinct patches does not enhance the maximum sustainable yield (MSY), it can indeed augment the maximum economic yield (MEY), particularly under conditions of asymmetric migration. This notable result emerges from complex interactions among local growth dynamics, harvest pressures, and rapid dispersal rates, thus creating a slow–fast dynamical system conducive to rigorous mathematical exploration.

Spatial heterogeneity and inter-patch migration are essential considerations in resource management and conservation biology. Marine populations frequently occupy discrete habitat patches—such as distinct reefs or fishing zones—where migration significantly influences population stability and harvest outcomes [9]. Moreover, climate change introduces additional complexity, altering species migration patterns and habitat connectivity through changing ocean temperatures [10].

This paper offers a detailed mathematical analysis of a two-patch fish population model incorporating logistic growth within each patch, a constant harvesting effort, and migration modulated by a temperature-dependent function, $\gamma(T)$. Two-patch models have extensively contributed to understanding fish populations moving between areas with differing ecological and managerial conditions [14, 15]. Prior studies have shown that dispersal rates significantly influence population stability, particularly under differential fishing pressures [4, 14, 15]. Specifically, uneven exploitation in interconnected patches may lead to intricate stability outcomes, potentially causing instabilities if migration rates are high and management is uncoordinated

[15, 14]. Conversely, coordinated patch management has the potential to significantly enhance overall yields, as demonstrated by Auger et al. [4], who found that high dispersal can increase total fishery yield beyond the isolated management yields.

The primary objective of this research is to analytically investigate a two-patch logistic fishery model wherein migration asymmetry is regulated by the temperature-dependent parameter $\gamma(T)$. This approach enables the exploration of how environmental temperature changes may affect migration dynamics, potentially driving directional movements such as fish seeking cooler habitats [10]. Specifically, this study aims to:

(i) Analyze how variations in $\gamma(T)$ influence net migration flow and equilibrium distributions between patches.

(ii) Determine equilibrium points for the system and conduct local stability analyses for scenarios where migration rate $D > 0$, identifying stable and center manifolds, particularly at bifurcation points.

(iii) Assess the impacts of critical parameters—migration rate $D$, fishing effort $E$, and temperature $T$—on fish populations within each patch.

(iv) Formulate and analyze yield and profit functions, explicitly deriving sustainable catch rates and optimal harvesting conditions.

(v) Establish conditions under which allowing migration between patches enhances yields and profits compared to isolated patch management.

Our methodology relies on analytical and Numerical approaches derived from differential equations and stability theory, ensuring rigorous and transparent results. Additionally, conceptual illustrations clarify key phenomena such as equilibrium bifurcations and relationships between yield and fishing effort, thereby facilitating interpretation. Finally, we complement our analytical findings with numerical simulations, illustrating transient behaviors and quantifying the effects of asymmetric migration on system trajectories.

# 2 Mathematical Model

We consider a two-patch fishery model with asymmetric migration,[5]

$$\frac{dx_1}{dt} = r_1 x_1 \left(1 - \frac{x_1}{K_1}\right) - E x_1 + D(\gamma(T)x_2 - x_1), \tag{2.1}$$

$$\frac{dx_2}{dt} = r_2 x_2 \left(1 - \frac{x_2}{K_2}\right) - E x_2 + D(x_1 - \gamma(T)x_2). \tag{2.2}$$

Here,

$$\gamma(T) = \gamma_0 + aT \tag{2.3}$$

- Logistic Growth: In the absence of harvesting and migration, each patch's population grows according to a logistic law. Patch $i$ has intrinsic growth rate $r_i$ and carrying capacity $K_i$. The logistic term is $r_i x_i(1 - \frac{x_i}{K_i})$, which ensures that $x_i$ tends to $K_i$ as $t \to \infty$ if unharvested and isolated. This term provides density-dependent self-limitation of population growth.

- Harvesting (Fishing Effort): A constant fishing effort $E$ (assumed equal in both patches for this model) removes fish from each patch at a rate proportional to the local population. The term $Ex_i$ represents the per-capita harvest mortality in patch $i$. This corresponds to a proportional harvesting strategy (also called constant-effort harvesting) – for example, a fixed number of fishing boats operating continuously in each patch yields a harvest rate proportional to stock size. If $E$ is large relative to $r_i$, it can drive the population down or even to extinction. Throughout, $E$ has units of [1/time] corresponding to an instantaneous harvest rate.

- Migration Between Patches: Fish move between the two patches at rate $D > 0$, which we assume constant (not density-dependent, aside from being proportional to population sizes available to move). The term $D(\gamma x_2 - x_1)$ in the $\frac{dx_1}{dt}$ equation represents net migration into patch 1: fish arrive from patch 2 at rate $D\gamma x_2$ and leave patch 1 at rate $Dx_1$. Similarly, $D(x_1 - \gamma x_2)$ in $\frac{dx_2}{dt}$ is the net migration into patch 2. The factor $\gamma = \frac{\gamma_2}{\gamma_1}$ modulates the asymmetry of migration: if $\gamma = 1$, migration is symmetric (fish leave each patch at the same per-capita rate $D$ and move into the other patch symmetrically). If $\gamma \neq 1$, the migration is asymmetric. Specifically, $\gamma x_2$ vs. $x_1$ creates an imbalance: for example, if $\gamma > 1$, the inflow to patch 1 ($D\gamma x_2$) is larger (for a given $x_2$) than the inflow to patch 2 (which is $Dx_1$ for a given $x_1$), implying a bias of movement from patch 2 towards patch 1. Conversely, $\gamma < 1$ would bias movement from patch 1 towards patch 2.

- Temperature-Dependent Migration Coefficient: The asymmetry parameter $\gamma$ is not a constant but a function of environmental temperature $T$. We are given $\gamma(T) = \gamma_0 + aT$, where $\gamma_0$ and $a$ are constants. This formulation means that environmental changes (e.g. ocean temperature changes) can alter migration patterns. The sign of $a$ dictates the direction of change: if $a > 0$, higher temperature $P$ increases $\gamma$, which (for $T$ uniform across the region) tends to increase the relative migration flow from patch 2 to patch 1. If $a < 0$, higher $T$ decreases $\gamma$, favoring movement from patch 1 to patch 2. We assume $T$ is a parameter (a slowly varying environmental variable or a scenario parameter) and not a dynamic variable in the model. Thus, we can analyze the system for different fixed values of $T$ to understand how a climate scenario affects the fish dynamics.

It is useful to interpret $\gamma$ in terms of fish behavior. Assuming patch 1 as a cool water refuge and patch 2 as a warm habitat. If rising temperatures cause fish to seek cooler waters, one would expect more fish to migrate from patch 2 to patch 1 as $T$ increases – this corresponds to $a > 0$ (increasing $\gamma$ biases net flow toward patch 1). On the other hand, some species might expand into warmer areas when temperatures rise, which would correspond to $a < 0$ (higher $T$ causing a shift of fish from patch 1 to patch 2). Our model can capture either scenario through the sign of $a$. Empirical studies have shown that temperature changes can indeed affect connectivity: for example, in many marine species, especially during larval dispersal, connectivity tends to decrease with higher temperatures (due to faster larval development and shorter dispersal times). In adult fish, temperature shifts can alter migration timing and routes– for instance, fish may migrate earlier in warmer years or move to maintain their preferred thermal habitat. Thus, $\gamma(T)$ is a simplified way to encode such temperature-driven migration behavior into the model. The intrinsic growth rates, $r_1$ and $r_2$, represent the inherent ability of the fish population in each patch to grow in the absence of limiting factors. Variations in these rates between the patches could be indicative of differences in habitat quality or environmental conditions affecting reproduction and survival. The carrying capacities, $K_1$ and $K_2$, define the upper limits on the population sizes that each patch can sustain based on the availability of resources. Differences in carrying capacities suggest variations in the capacity of the two habitats to support the fish population.

Fishing effort, $E$, introduces a mortality factor, assumed to be proportional to the population size in each patch with a rate of $E$. This parameter represents the intensity of fishing activity and is a key factor in fisheries management. The dispersal rate, $D$, governs the movement of fish between the two patches. The terms $D(\gamma x_2 - x_1)$ and $D(x_1 - \gamma x_2)$ describe the net flow of individuals into patch 1 and patch 2, respectively, suggesting a density-dependent movement.To avoid the extinction of the fished species, it is necessary to choose a fishing effort which remains lower than the popu-lation growth rate, i.e $E < r$.

The patch preference parameter, $\gamma$, introduces a bias in the dispersal. If $\gamma > 1$, it indicates a preference for moving from patch 1 to patch 2, and a resistance to moving from patch 2 to patch 1. Conversely, if $\gamma < 1$, there is a preference for patch 1. When $\gamma = 1$, the dispersal is symmetric, driven solely by the difference in population densities. This parameter can reflect behavioral responses of the fish to perceived differences in habitat quality or resource availability. Another important analysis of this model is the profit and maximum yeild for this,[5] ,Based on Schaefer's model, Gordon [6] introduced an economic model where profits noted by $\Pi_i$ are estimated by assuming constant price per unit of yield($Y_i$) noted by p and constant cost per unit of fishing effort c ,then

$$Y_i(E) = Ex_i^*,$$
$$\Pi_i(E) = E(px_i^* - c)$$

# 3 Analysis

**Equilibrium Point Analysis:**

Equilibrium points of the system are found by setting the rates of change to zero:

$$r_1x_1\left(1-\frac{x_1}{K_1}\right)-Ex_1+D(\gamma x_2-x_1)=0,$$
$$r_2x_2\left(1-\frac{x_2}{K_2}\right)-Ex_2+D(x_1-\gamma x_2)=0. \tag{3.1}$$

**Case:1**–When the patches are not connected ,i.e D=0, Analytical Proof of Unique Positive Equilibrium Consider the following two-patch logistic dispersal system,

$$\frac{dx_1}{dt}=r_1x_1\left(1-\frac{x_1}{K_1}\right)-Ex_1, \tag{3.2}$$

$$\frac{dx_2}{dt}=r_2x_2\left(1-\frac{x_2}{K_2}\right)-Ex_2. \tag{3.3}$$

For species 1, setting $\frac{dx_1}{dt}=0$ gives

$$x_1^*\Big(r_1\Big(1-\frac{x_1^*}{K_1}\Big)-E\Big)=0.$$

In general for i disconnected patches,

- For isolated Patches $(D=0)$

$$\frac{dx_i}{dt}=r_ix_i\Big(1-\frac{x_i}{K_i}\Big)-Ex_i$$

  - Finding out the equilibrium point/biomass

$$x_i=x_i^*=K_i\left(1-\frac{E}{r_i}\right), Positive\ when\ E<r_i\ ,$$
$$and\ x_i=0.$$

So,the equilibrium points are $E_0(0,0)$ *and* $E_1\left(K_1(1-\frac{E}{r_1}),K_2(1-\frac{E}{r_2})\right)$

Now,for stability check the Jacobian Matrix at (0,0)is,

$$J(x_1,x_2)=\begin{pmatrix} r_1-E & 0 \\ 0 & r_2-E \end{pmatrix}.$$

So,the eigen values are,$\lambda_1=r_1-E\ and\ \lambda_2=r_2-E$ ,since $E<r_i$ both the eigen values are real and positive,implies the equilibrium point is unstable.

Now,the Jacobian matrix is at $E_1$ is,

$$J(x_1,x_2)=\begin{pmatrix} E-r_1 & 0 \\ 0 & E-r_2 \end{pmatrix}.$$

So,the eigen values are,$\lambda_1=E-r_1\ and\ \lambda_2=E-r_2$ ,since $E<r_i$ both the eigen values are real and negative,implies the equilibrium point locally asymptotically stable.

Now,for the global stability ,using standard logistic–harvest Lyapunov function

$$V=x_i-x_i^*-x_i^*ln\frac{x_i}{x_i^*}$$
$$and\ \dot{V}=\frac{dV}{dx_1}\dot{x}_1+\frac{dV}{dx_2}\dot{x}_2$$
$$=-\frac{r_1}{K_1}(x_1-x_1^*)^2-\frac{r_2}{K_2}(x_2-x_2^*)^2\leq 0$$

i,e the positive equilibrium $(x_1^*,x_2^*)$ is globally asymptotically stable on the interior of the positive quadrant.

- **Maximum Yield**

  The yield for species 1 is

$$Y_1 = Ex_1^* = EK_1\Big(1 - \frac{E}{r_1}\Big),$$

  and for species 2 is

$$Y_2 = Ex_2^* = EK_2\Big(1 - \frac{E}{r_2}\Big).$$

  To find the fishing effort $E$ that maximizes the yield for species 1, differentiate $Y_1$ with respect to $E$:

$$\frac{dY_1}{dE} = K_1\Big(1 - \frac{E}{r_1}\Big) - E\,K_1\Big(\frac{1}{r_1}\Big) = K_1\Big(1 - \frac{2E}{r_1}\Big).$$

  Setting $\frac{dY_1}{dE} = 0$ yields

$$1 - \frac{2E}{r_1} = 0 \quad \Longrightarrow \quad E = \frac{r_1}{2}.$$

  Similarly, for species 2 the maximum yield occurs at $E = \frac{r_2}{2}$. Hence, the maximum yields are

$$Y_{1,\max} = \frac{r_1}{2}K_1\left(1 - \frac{1}{2}\right) = \frac{r_1K_1}{4}, \quad Y_{2,\max} = \frac{r_2K_2}{4}.$$

- **Economic Profit**

  The profit for species 1 is given by

$$\Gamma_1 = \big(p\,x_1^* - c\big)E = \Big(pK_1\Big(1 - \frac{E}{r_1}\Big) - c\Big)E,$$

  and similarly for species 2,

$$\Gamma_2 = \Big(pK_2\Big(1 - \frac{E}{r_2}\Big) - c\Big)E.$$

  To maximize profit for species 1, differentiate $\Gamma_1$ with respect to $E$,

$$\frac{d\Gamma_1}{dE} = pK_1\Big(1 - \frac{E}{r_1}\Big) - \frac{pK_1}{r_1}E - c.$$

$$pK_1 - \frac{2pK_1}{r_1}E - c = 0 \quad \Longrightarrow \quad E = \frac{r_1}{2}\Big(1 - \frac{c}{pK_1}\Big).$$

  Thus, the optimal fishing effort for species 1 is

$$E_1^* = \frac{r_1}{2}\Big(1 - \frac{c}{pK_1}\Big),$$

  provided $pK_1 > c$ and $E_1^* < r_1$. Similarly, for species 2, the optimal effort is

$$E_2^* = \frac{r_2}{2}\Big(1 - \frac{c}{pK_2}\Big).$$

  The maximum profits are obtained by substituting these optimal efforts back into the respective profit equations. The price p and cost c significantly influence the optimal harvesting rates for maximizing profit.

Dispersal parameters and asymmetry significantly influence yield and profit. An increase in dispersal generally leads to resource redistribution, potentially enhancing yield in less productive patches. However, excessive dispersal may reduce overall equilibrium populations, negatively affecting yield and profit.

Specifically,

- Increasing equilibrates resource levels across patches, improving yield in the less productive patch but possibly decreasing total yield.

- Asymmetry parameter influences directional dispersal. Higher means preferential dispersal toward the first patch, benefiting its yield and profit if it's economically advantageous.

Thus, optimal profit and yield depend critically on appropriately balancing dispersal intensity and asymmetry .

**Case-2**:When $D > 0$,i,e moderate dispersal. Setting $\frac{dx_1}{dt} = 0$ and $\frac{dx_2}{dt} = 0$, we have the equilibrium equations:

**Theorem 3.1.** *From the system (3.1) we say,*

*(a) There exists a trivial unstable equilibrium,* $\hat{E} = (0, 0)$.

*(b) There exists a unique positive equilibrium,* $E^* = (x_1^*, x_2^*)$, *which is locally asymptotically stable.*

*(c)* $E^*$ *is globally asymptotically stable with respect to* $\mathbb{R}_+^2 \setminus \{(0, 0)\}$.

*Proof.* Consider the following two-patch logistic system,

$$\begin{cases} \dfrac{dx_1}{dt} = r_1\, x_1\Big(1 - \frac{x_1}{K_1}\Big) \;-\; E\, x_1 \;+\; D\,\big(\gamma\, x_2 - x_1\big), \\ \dfrac{dx_2}{dt} = r_2\, x_2\Big(1 - \frac{x_2}{K_2}\Big) \;-\; E\, x_2 \;+\; D\,\big(x_1 - \gamma\, x_2\big), \end{cases}$$

where all parameters are positive constants,

$$r_1, r_2 > 0, \quad K_1, K_2 > 0, \quad D > 0, \quad \gamma > 0, \quad E > 0.$$

We wish to show that, under suitable "ecological" conditions such as

$$r_1 > E + D \quad \text{and} \quad r_2 > E + D\,\gamma,$$

there exists a *unique* equilibrium point $\big(x_1^*, x_2^*\big)$ with $x_1^* > 0$ and $x_2^* > 0$.

At equilibrium, we require,

$$\begin{cases} 0 \;=\; r_1\, x_1\Big(1 - \frac{x_1}{K_1}\Big) \;-\; E\, x_1 \;+\; D\big(\gamma\, x_2 - x_1\big), \\ 0 \;=\; r_2\, x_2\Big(1 - \frac{x_2}{K_2}\Big) \;-\; E\, x_2 \;+\; D\big(x_1 - \gamma\, x_2\big). \end{cases}$$

from here,we get

**First equation: Solve for $x_2$.**

$$r_1\, x_1\Big(1 - \frac{x_1}{K_1}\Big) \;-\; E\, x_1 \;+\; D\big(\gamma x_2 - x_1\big) = 0 \implies D\,\gamma\, x_2 = \frac{r_1}{K_1}x_1^2 + (E + D - r_1)x_1.$$

Hence

$$x_2 \;=\; g_1(x_1) \;=\; \frac{1}{D\,\gamma}\Big[\frac{r_1}{K_1}x_1^2 + (E + D - r_1)\, x_1\Big].$$

**Second equation: Solve for $x_1$.**

$$r_2\, x_2\Big(1 - \frac{x_2}{K_2}\Big) \;-\; E\, x_2 \;+\; D(x_1 - \gamma x_2) = 0 \implies D\, x_1 = \frac{r_2}{K_2}x_2^2 \;+\; (E + D\,\gamma - r_2)\, x_2.$$

Hence

$$x_1 \;=\; g_2(x_2) \;=\; \frac{1}{D}\Big[\frac{r_2}{K_2}\, x_2^2 + (\, E + D\,\gamma - r_2\,)\, x_2\Big].$$

Defining the *composite function*

$$h(x_1) \;=\; g_2\big(g_1(x_1)\big).$$

A positive equilibrium $\big(x_1^*, x_2^*\big)$ must satisfy

$$x_2^* \;=\; g_1(x_1^*), \quad x_1^* \;=\; g_2(x_2^*).$$

Hence $x_1^*$ must solve

$$h(x_1) \;=\; x_1.$$

Once we find $x_1^* > 0$ with $h(x_1^*) = x_1^*$, we set $x_2^* = g_1(x_1^*) > 0$.

**Continuity.** Both $g_1(x_1)$ and $g_2(x_2)$ are polynomial expressions in $x_1$ or $x_2$. Therefore they are continuous for $x_1, x_2 \geq 0$. Consequently, $h(x_1) = g_2\big(g_1(x_1)\big)$ is also continuous on $\mathbb{R}_+^2$.

**Behavior at $0$ and $\infty$.**

- *At $x_1 = 0$:*

  $$g_1(0) = 0 \implies g_2\big(g_1(0)\big) = g_2(0) = 0,$$

  so

  $$h(0) = 0.$$

  Thus $h(0) = 0$, corresponding to the trivial (boundary) equilibrium $(0, 0)$.

- *For large $x_1$:* As $x_1 \to \infty$, the term $\frac{r_1}{K_1} x_1^2$ dominates $g_1(x_1)$, so $g_1(x_1) \sim \frac{r_1}{D\gamma K_1} x_1^2$. Then $g_2\big(g_1(x_1)\big)$ is roughly proportional to $\big(g_1(x_1)\big)^2 \sim x_1^4$. Hence

  $$h(x_1) \;=\; g_2\big(g_1(x_1)\big) \sim (\text{const}) \times x_1^4 \implies h(x_1) > x_1 \quad \text{for sufficiently large } x_1.$$

**Applying the Intermediate Value Theorem.** Since $h(0) = 0$ and $h(x_1) > x_1$ for large $x_1$, we see:

$$\text{At } x_1 = 0, \quad h(0) = 0 < x_1 = 0 \text{ (they coincide, trivial eqn).}$$

$$\text{For sufficiently large } x_1, \quad h(x_1) > x_1.$$

By continuity, there must exist an $x_1^* > 0$ such that $h(x_1^*) = x_1^*$. This gives a *positive* solution, which in turn yields a positive equilibrium $(x_1^*, x_2^*)$ where $x_2^* = g_1(x_1^*) > 0$.

To ensure *only one* such intersection occurs, we show $h(x_1)$ is strictly increasing, so $h$ can meet the line $y = x$ at most once.

So,

$$h'(x_1) = \frac{d}{dx_1}\Big[\, g_2\big(g_1(x_1)\big)\Big] = g_2'\big(g_1(x_1)\big) \;\cdot\; g_1'(x_1).$$

Next, we compute $g_1'(x_1)$ and $g_2'(x_2)$,

$$g_1'(x_1) = \frac{1}{D\gamma}\Big[2\,\frac{r_1}{K_1}\,x_1 + (\,E + D - r_1\,)\Big],$$

$$g_2'(x_2) = \frac{1}{D}\Big[2\,\frac{r_2}{K_2}\,x_2 + (\,E + D\,\gamma - r_2\,)\Big].$$

**Positivity of these derivatives.** Under typical ecological assumptions $r_1 > E+D$ and $r_2 > E+D\gamma$, the constants $(E+D-r_1)$ and $(E+D\gamma-r_2)$ are negative but can be overridden by the linear (or quadratic) terms in $x_1$ and $x_2$ as soon as $x_1, x_2 > 0$ are not too small.

- Whenever

$$x_1 > \frac{(r_1 - E - D)\,K_1}{2\,r_1},$$

the expression $2\,\frac{r_1}{K_1}\,x_1 + (E+D-r_1)$ is $> 0$, so $g_1'(x_1) > 0$.

Likewise, if

$$x_2 > \frac{(r_2 - E - D\gamma)\,K_2}{2\,r_2},$$

then $2\,\frac{r_2}{K_2}\,x_2 + (E+D\gamma-r_2) > 0$, so $g_2'(x_2) > 0$.

Because $g_1(x_1)$ itself becomes positive for $x_1 > 0$ large enough, $g_2'\big(g_1(x_1)\big) > 0$ under these conditions. Thus for $x_1 > 0$ in the relevant range:

$$h'(x_1) = g_2'\big(g_1(x_1)\big)\ \cdot\ g_1'(x_1)\ >\ 0.$$

Now,If $h$ is strictly increasing, then the equation $h(x_1) = x_1$ can have *at most* one solution. Indeed, suppose there were two solutions $x_1' < x_1''$ with $h(x_1') = x_1'$ and $h(x_1'') = x_1''$. Then strict increase would imply

$$h(x_1'')\ >\ h(x_1')\ \implies\ x_1''\ >\ x_1',$$

yet this conflicts with $h(x_1'') = x_1''$ and $h(x_1') = x_1'$. Hence only one intersection is possible. So,combining this uniqueness with the existence result from the Intermediate Value Theorem, we conclude: There is at least one positive root of $h(x_1) = x_1$. strict monotonicity ($h'(x_1) > 0$) forbids more than one positive root. So,when,$h(0) = 0$ and for sufficiently large $x_1^*$,$h(x_1^*) < 0$. Thus, this curve is concave downward and crosses from positive to negative exactly once, producing a unique positive intercept along the axes.

Hence, the positive quadrant portion of curve A is strictly decreasing after reaching a maximum, implying monotone decrease after a positive peak. Notice $g(0) = 0$ and for sufficiently large $x_2^*$,$g(x_2^*) < 0$. Thus, this curve is concave downward and crosses from positive to negative exactly once, producing a unique positive intercept along the axes.

Hence, the positive quadrant portion of curve A is strictly decreasing after reaching a maximum, implying monotone decrease after a positive peak. Both functions $h(x_1^*)$ and $g(x_2^*)$ are concave downward, crossing axes exactly once, and monotonically decreasing after their maxima. Due to the monotonic decrease and smooth differentiability, the curves can intersect at most once in the positive quadrant, establishing uniqueness of the positive equilibrium. Therefore there is **exactly one** positive equilibrium $\big(x_1^*, x_2^*\big)$ with $x_1^* > 0, x_2^* > 0$.

**Stability Analysis**

The local stability of an equilibrium point $(x_1^*, x_2^*)$ is determined by the eigenvalues of the Jacobian matrix evaluated at that point. The Jacobian matrix of the system is given by,

$$J(x_1, x_2) = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}.$$

$$\begin{aligned}
J_{11} &= \frac{\partial}{\partial x_1}\left(r_1 x_1\left(1 - \frac{x_1}{K_1}\right) - E x_1 + D(\gamma x_2 - x_1)\right)\\
J_{12} &= \frac{\partial}{\partial x_2}\left(r_1 x_1\left(1 - \frac{x_1}{K_1}\right) - E x_1 + D(\gamma x_2 - x_1)\right)\\
J_{21} &= \frac{\partial}{\partial x_1}\left(r_2 x_2\left(1 - \frac{x_2}{K_2}\right) - E x_2 + D(x_1 - \gamma x_2)\right)\\
J_{22} &= \frac{\partial}{\partial x_2}\left(r_2 x_2\left(1 - \frac{x_2}{K_2}\right) - E x_2 + D(x_1 - \gamma x_2)\right)
\end{aligned}$$

Then,

$$J(x_1,x_2)=\begin{pmatrix} r_1-\frac{2r_1}{K_1}x_1-E-D & D\gamma \\ D & r_2-\frac{2r_2}{K_2}x_2-E-D\gamma \end{pmatrix}.$$

**Stability of the Trivial Equilibrium** $(0,0)$

Evaluating the Jacobian at $(0,0)$, we get,

$$J(0,0)=\begin{pmatrix} r_1-E-D & D\gamma \\ D & r_2-E-D\gamma \end{pmatrix}.$$

The stability is determined by the eigenvalues of this matrix, which are the roots of the characteristic equation $\det(J(0,0)-\lambda I)=0$,

$$(r_1-E-D-\lambda)(r_2-E-D\gamma-\lambda)-(D\gamma)(D)=0.$$

Expanding, we have,

$$\lambda^2-\lambda(r_1-E-D+r_2-E-D\gamma)+(r_1-E-D)(r_2-E-D\gamma)-D^2\gamma=0.$$

The trivial equilibrium is locally asymptotically stable if both eigenvalues have negative real parts. This occurs if and only if the trace of the Jacobian is negative and the determinant is positive.

$$\mathrm{Tr}(J(0,0))=r_1+r_2-D\gamma-2E-D,$$

$$\begin{aligned}\det(J(0,0)) &= r_1r_2-r_1E-r_1D\gamma-r_2E+E^2+ED\gamma-r_2D+DE+D^2\gamma-D^2\gamma.\\ &= E^2+ED\gamma+DE+r_1(r_2-E-D\gamma)-r_2(E+D)\\ \Delta &= (trace)^2-4(determinant)=D^2(\gamma+1)^2+2(r_1-r_2)(\gamma-1)D+(r_1-r_2)^2\end{aligned}$$

For extinction to be stable, we require both eigenvalues to have $\Re(\lambda)<0$. If either becomes positive, extinction is unstable (meaning populations can grow from small introduction). Let, $r_1>E+D$ , and $r_2>E+D\gamma$ The conditions for stability of the trivial equilibrium are,

$$\begin{aligned} r_1+r_2-D\gamma-2E-D &> 0,\\ (r_1-E-D)(r_2-E-D\gamma)-D^2\gamma &< 0.\\ \Delta &> 0\end{aligned}$$

For $r_1>E+D$, $trace(J_0)>0, determinant(J_0)<0, \Delta>0$, which makes $E_0(0,0)$ unstable. So,the trivial equilibrium is unstable, suggesting the population growth from low densities.

**Stability of Coexistence Equilibria**

For the coexistence equilibria $(x_1^*,x_2^*)$, the stability is determined by the eigenvalues of the Jacobian matrix evaluated at these points,

$$J(x_1^*,x_2^*)=\begin{pmatrix} r_1-\frac{2r_1}{K_1}x_1^*-E-D & D\gamma \\ D & r_2-\frac{2r_2}{K_2}x_2^*-E-D\gamma \end{pmatrix}.$$

The determinant is,

$$\begin{aligned} det(J^*) \quad &= \left(r_1-\frac{2r_1}{K_1}x_1^*-E-D\right)\left(r_2-\frac{2r_2}{K_2}x_2^*-E-D\gamma\right)-D^2\gamma\\ \implies det(J^*) \quad &= DE\gamma-D\gamma r_1+\frac{2D\gamma x_1^*r_1}{K_1}+DE-Dr_2-\frac{2Dx_2^*r_2}{K_2}+E^2-Er_1-Er_2\\ &+\frac{2Ex_2^*r_2}{K_2}+\frac{2Ex_1^*r_1}{K_2}+r_1r_2-\frac{2x_2^*r_1r_2}{K_2}-\frac{2K_2x_1^*r_1r_2}{K_1}+\frac{4x_1^*x_2^*r_1r_2}{K_1K_2}>0\end{aligned}$$

And,the trace is,

$$\begin{aligned}
trace(J^*) &= \left(r_1 - \frac{2r_1}{K_1}x_1^* - E - D + r_2 - \frac{2r_2}{K_2}x_2^* - E - D\gamma\right)\\
\implies trace(J) &= r_1 + r_2 - 2E - D - D\gamma - (\frac{2r_1}{K_1}x_1^* + \frac{2r_2}{K_2}x_2^*) < 0\\
And, \Delta \;\; = & \;\; \frac{1}{K_1^2K_2^2}\Bigg[4K_2^2x_1^2r_1^2 - 4x_1^*r_1K_1K_2\Bigg\{2r_2x_2 + K_2\{r_1 - r_2 + D(\gamma-1)\}\Bigg\} + \{r_1^2 + (-2r_2 + (2\gamma-2)D)r_1\\
& +r_2^2 - 2D(\gamma-1)r_2 + D^2(\gamma+1)^2\}K_2^2 + 4r_2x_2^*\{r_1 - r_2 + D(\gamma-1)\}K_2 + 4x_2^2r_2^2)K_1^2\Bigg] > 0
\end{aligned}$$

Since ,$\Delta > 0$ and $trace(J^*) < 0$ ,so $E^*(x_1^*, x_2^*)$ is locally aymptotically stable. For determining the global stability of $E^*$ we assume the function $\phi(x_1, x_2) = \frac{1}{x_1x_2}$,for $(x_1, x_2) \in \mathbb{R}^2$ .Then,we get,

$$\begin{aligned}
\frac{\partial}{\partial x_1}\left(\phi(x_1,x_2)f(x_1,x_2)\right) &= \frac{\partial}{\partial x_1}(\frac{1}{x_1x_2})\left[x_1r_1\left(1 - \frac{x_1}{K_1}\right) + D(\gamma x_2 - x_1) - Ex_1\right],\\
\frac{\partial}{\partial x_2}\left(\phi(x_1,x_2)g(x_1,x_2)\right) &= \frac{\partial}{\partial x_2}(\frac{1}{x_1x_2})\left[x_2r_2\left(1 - \frac{x_2}{K_2}\right) + D(x_1 - \gamma x_2) - Ex_2\right].
\end{aligned} \tag{3.4}$$

So,

$$\begin{aligned}
\frac{\partial}{\partial x_1}\left(\phi(x_1,x_2)f(x_1,x_2)\right) &= \frac{\partial}{\partial x_1}\left[\frac{r_1}{x_2}\left(1 - \frac{x_1}{K_1}\right) + \frac{D\gamma}{x_1} - \frac{D}{x_2} - \frac{E}{x_2}\right],\\
&= -\frac{r_1}{x_2K_1} - \frac{D\gamma}{x_1^2} \leq 0.
\end{aligned}$$

And,

$$\begin{aligned}
\frac{\partial}{\partial x_2}\left(\phi(x_1,x_2)g(x_1,x_2)\right) &= \frac{\partial}{\partial x_2}\left[\frac{r_2}{x_1}\left(1 - \frac{x_2}{K_2}\right) - \frac{D\gamma}{x_1} + \frac{D}{x_2} - \frac{E}{x_1}\right],\\
&= -\frac{r_2}{x_1K_2} - \frac{D}{x_2^2} \leq 0.
\end{aligned}$$

Combining terms:

$$\frac{\partial}{\partial x_1}\left(\phi(x_1,x_2)f(x_1,x_2)\right) + \frac{\partial}{\partial x_2}\left(\phi(x_1,x_2)g(x_1,x_2)\right) \leq 0$$

Therefore,by Dulac's Criterion,we can conclude that there will be no closed orbits in $\mathbb{R}^2$.According to the Poincare-Bendixon Theorem for 2-dimensional systems,any solution in the closed region bounded by $N(x_1, x_2) = n^*$ must converge to an equilibrium point or closed orbit.Since $E^*$ is the only stable equilibrium in this region,and we have shown that there are no closed orbits,all solutions must converge to $E^*$.Therefore,$E^*$ is globally asymptotically stable. □

# 4 Case-3:When $D \to \infty$ or for fast migration

When there is first migration among then patches ,i.e $D \to \infty$ the the model comes by using singular perturbation theory, Tikhonov's theorem ([12]; [13]; [11]) and Elbetch [8]( Theorem 3.5) we determine the reduced model of (2.3), Let,W(t) is the solution of logistic equation then,

$$\frac{dW}{dt} = RX(1 - \frac{X}{K}) - EX$$

Where,

- W is the total biomass=$x_1+x_2$=X
- R,K -effective growth rate and carrying capacity that incorporate migration and patch heterogeneity

Where,$K=(\gamma+1)\frac{\gamma r_1+r_2}{\gamma^2\frac{r_1}{K_1}+\frac{r_2}{K_2}}$, $R=\frac{\gamma r_1+r_2}{\gamma+1}$

The $D\to\infty$ case is interesting because it assumes very fast movements of fish between sites in relation to slow stock growth and fishing, which is relatively realistic. In addition, this hypothesis makes it possible to derive a reduced model which is simpler and easier to analyze. The case D = 0 is a bit trivial in our case because it corresponds to isolated sites with an excess yield always zero. The case with any D is of course more general and would deserve a more in-depth study later. Let us recall some results about the single-species logistic equation with harvesting (2.3). We are going to choose the harvesting effort E in order to maximize the total profit. The logistic growth equation (1.1) with constant harvesting has two equilibria $x=0$, and $x^*(E):=K(1-\frac{E}{r})$, being positive when $E<r$. When $x^*(E)>0$, it is stable and the equilibrium $x^*=0$ is unstable and otherwise $x^*=0$ is stable.Now,

$$\Pi(E):=(px-c)E. \tag{4.1}$$

At equilibrium, where

$$x^*(E)=K\left(1-\frac{E}{r}\right),$$

we obtain,

$$\Pi^*(E):=(px^*(E)-c)E=-\frac{pK}{r}E^2+(pK-c)E. \tag{4.2}$$

This expression is positive for all

$$E<r\left(1-\frac{c}{pK}\right).$$

The profit function $\Pi^*(E)$ is a downward-opening parabola with respect to the fishing effort $E$, attaining a maximum at,

$$E^*_{\mathrm{MEY}}=\begin{cases}\dfrac{(pK-c)}{2pK}r & \text{if } c<pK,\\ 0 & \text{otherwise.}\end{cases}$$

The corresponding value of the stable, non-trivial positive equilibrium $x^*$ is,

$$x^*_{\mathrm{MEY}}:=x^*(E^*_{\mathrm{MEY}})=\begin{cases}\dfrac{K}{2}\left(1-\dfrac{c}{pK}\right) & \text{if } c<pK,\\ 0 & \text{otherwise.}\end{cases} \tag{4.3}$$

The corresponding profit at MEY, denoted by $\Pi^*_{\mathrm{MEY}}$, is obtained by substituting $E^*_{\mathrm{MEY}}$ into $\Pi^*(E)$,

$$\Pi^*_{\mathrm{MEY}}=(pK-c)E^*_{\mathrm{MEY}}-\frac{pK}{r}(E^*_{\mathrm{MEY}})^2. \tag{4.4}$$

After substituting $E^*_{\mathrm{MEY}}$ into equation(4.4), we obtain,

$$\Pi^*_{\mathrm{MEY}}=\frac{(pK-c)^2}{4pK}r.$$

We assume that equation (2.6) admits a unique positive equilibrium, denoted by $E^*_n(D;E)$, which depends on the migration rate $D$ and fishing effort $E$. At equilibrium, the profit $\Pi^*_i(D;E)$ of patch $i$ as a function of $D$ and fishing effort is given by the following formula,

$$\Pi^*_i(D;E)=(p_ix^*_i(D;E)-c_i)E,\quad i=1,\dots,n. \tag{4.5}$$

For $D = 0$, we obtain,

$$\begin{aligned}\Pi_i^*(0;E) &:= (p_i x_i^*(0;E) - c_i)E \\ &= \left(p_i K_i \left(1 - \frac{E}{r_i}\right) - c_i\right) E \\ &= (p_i K_i - c_i)E - \frac{p_i K_i}{r_i} E^2.\end{aligned}$$

The profit $\Pi_i^*(D;E)$ of patch $i$ is a downward-opening (negative) parabola with respect to the fishing effort $E$, attaining a maximum at,

$$E_{\mathrm{MEY},i}^{*,0} = \begin{cases} \dfrac{(p_i K_i - c_i)}{2 p_i K_i} r_i & \text{if } c_i < p_i K_i, \\ 0 & \text{otherwise.} \end{cases}$$

In general, the values $E_{\mathrm{MEY},i}^{*,0}$ differ across patches $i$ because they depend on the parameters of each patch.

The corresponding profit at MEY of patch $i$, denoted by $\Pi_{\mathrm{MEY},i}^{*,0}$, is obtained by replacing the fishing effort at MEY, $E_{\mathrm{MEY},i}^{*,0}$, in $\Pi_i^*(0;E)$. Thus, we have the expression,

$$\Pi_{\mathrm{MEY},i}^{*,0} = (p_i K_i - c_i) E_{\mathrm{MEY},i}^{*,0} - \frac{p_i K_i}{r_i} (E_{\mathrm{MEY},i}^{*,0})^2. \tag{4.6}$$

After substituting $E_{\mathrm{MEY},i}^{*,0}$ into equation (4.6), we get for all $i$,

$$\Pi_{\mathrm{MEY},i}^{*,0} = \frac{(p_i K_i - c_i)^2}{4 p_i K_i} r_i. \tag{4.7}$$

The sum of $\Pi_{\mathrm{MEY},i}^{*,0}$ over all patches $i$ gives the total profit of $n$ isolated patches,

$$\Pi_{\mathrm{MEY},T}^{*,0} = \sum_{i=1}^{n} \Pi_{\mathrm{MEY},i}^{*,0} = \sum_{i=1}^{n} \frac{(p_i K_i - c_i)^2}{4 p_i K_i} r_i. \tag{4.8}$$

When $D \to \infty$, we obtain,

$$\Pi_T^*(\infty;E) := \sum_{i=1}^{n} (p_i x_i^*(\infty;E) - c_i)E = \sum_{i=1}^{n} \left(p_i K \left(1 - \frac{E}{R}\right) - c_i\right) E. \tag{4.9}$$

This simplifies to,

$$\Pi_T^*(\infty;E) = \left(\sum_{i=1}^{n} p_i \; K - \sum_{i=1}^{n} c_i\right) E - \left(\sum_{i=1}^{n} p_i\right) \frac{K}{R} E^2. \tag{4.10}$$

Thus, the total profit simplifies to,

$$\Pi_T^*(\infty;E) = (PK - C)E - P\frac{K}{R}E^2, \tag{4.11}$$

we define,

$$C := c, \quad P := p,$$

The total profit $\Pi_T^*(\infty;E)$ is a downward-opening parabola in the fishing effort $E$, attaining a maximum at:

$$E_{\mathrm{MEY},T}^{*,\infty} = \begin{cases} \dfrac{PK - C}{2PK} R & \text{if } C < PK, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding total profit at MEY, $\Pi^{*,\infty}_{\mathrm{MEY},T}$, is obtained by substituting $E^{*,\infty}_{\mathrm{MEY},T}$ into $\Pi^{*}_{T}(\infty;E)$,

$$\Pi^{*,\infty}_{\mathrm{MEY},T} = (PK-C)E^{*,\infty}_{\mathrm{MEY},T} - P\frac{K}{R}(E^{*,\infty}_{\mathrm{MEY},T})^2. \tag{4.12}$$

Hence, we get:

$$\Pi^{*,\infty}_{\mathrm{MEY},T} = \frac{(PK-C)^2}{4PK}R.$$

Now,from Bilel Elbetch, Ali Moussaoui, Pierre Auger(2023)[7] we get thet,the excess profit at MEY $\Pi^{*}_{\mathrm{MEY}}$ is positive if and only if, the following condition hold,

$$\Pi_{\infty} > \Pi_{E_1} + \Pi_{E_2}$$
$$\implies R\frac{(PK-C)^2}{4PK} > r_1\frac{(pK_1-c)^2}{4pK_1} + r_2\frac{(pK_2-c)^2}{4pK_2}$$

From here,I will derive a condition on assymetry $\gamma$ ,on which price and cost depends,when $K_1 = K_2, r_1 = r_2 = r$

$$R\frac{(PK-C)^2}{4PK} > r_1\frac{(pK_1-c)^2}{4pK_1} + r_2\frac{(pK_2-c)^2}{4pK_2}$$
$$R\frac{(pK-C)^2}{4pK} > \frac{2r(pK_1-c)^2}{4pK_1}$$

substitute the value of R and K,

$$\frac{(1+\gamma)r}{\gamma+1}\frac{\{\frac{pr(1+\gamma)^2}{\frac{r}{K_1}(\gamma^2+1)} - c\}^2}{4p\frac{r(1+\gamma)^2}{\frac{r}{K_1}(\gamma^2+1)}} > \frac{r(pK_1-c)^2}{2pK_1}$$
$$\frac{r\{\frac{pK_1(1+\gamma)^2}{(\gamma^2+1)} - c\}^2}{4pK_1\frac{(1+\gamma)^2}{(\gamma^2+1)}} > \frac{r(pK_1-c)^2}{2pK_1}$$
$$\left(\frac{pK_1(1+\gamma)^2 - c(1+\gamma^2)}{(1+\gamma^2)}\right)^2 > 2r(pK_1-c)^2\frac{(1+\gamma)^2}{(1+\gamma^2)}$$
$$1 - \frac{c}{pK_1} > \frac{2\gamma}{1+\gamma^2 - 2r(1+\gamma)\sqrt{1+\gamma^2}}$$

So,the condition of profit of connected patch is $>$ the sum of the profit of isolated profit as long as this condition satisfies. if $\gamma$ is small enough then we can say,$c < pK_1$

# 5 Critical Fishing Effort

Now,from the equilibrium equation,

$$\begin{cases} 0 \;=\; r_1\,x_1^*\Big(1-\frac{x_1^*}{K_1}\Big) \;-\; E\,x_1^* \;+\; D\big(\gamma\,x_2^* - x_1^*\big), \\ 0 \;=\; r_2\,x_2^*\Big(1-\frac{x_2^*}{K_2}\Big) \;-\; E\,x_2^* \;+\; D\big(x_1^* - \gamma\,x_2^*\big). \end{cases}$$

From above equation, assuming $x_1^*, x_2^* > 0$, we divide the first by $x_1^*$ and the second by $x_2^*$,

$$x_1^*\left(r_1(1-\frac{x_1^*}{K_1}) - E - D\right) + D\gamma x_2^* = 0$$
$$\implies D\frac{\gamma x_2^*}{x_1^*} - D = E - r_1\left(1-\frac{x_1^*}{K_1}\right)$$
$$Similarly, D\frac{x_1^*}{x_2^*} - D\gamma = E - r_2\left(1-\frac{x_2^*}{K_1}\right)$$

Simplifying,

$$\begin{aligned} D\left(\frac{\gamma x_2^*}{x_1^*} - 1\right) &= E - r_1\left(1 - \tfrac{x_1^*}{K_1}\right), \\ D\left(\frac{x_1^*}{x_2^*} - \gamma\right) &= E - r_2\left(1 - \tfrac{x_2^*}{K_2}\right). \end{aligned}$$

Let $R = \frac{x_1^*}{x_2^*}$ denote the population ratio. Then, the equations become:

$$D\left(\frac{\gamma}{R} - 1\right) = E - r_1\left(1 - \frac{x_1^*}{K_1}\right),$$
$$D(R - \gamma) = E - r_2\left(1 - \frac{x_2^*}{K_2}\right).$$

At equilibrium, the left-hand sides are negatives of each other, so,

$$D\left(\frac{\gamma}{R} - 1\right) = -D(R - \gamma).$$

This identity simplifies to,

$$\frac{\gamma}{R} + R = \gamma + 1,$$

which holds if and only if $R = \gamma$, i.e., $x_1^* = \gamma x_2^*$.

Substitute $R = \gamma$ into the first equation,

$$D\left(\frac{\gamma}{\gamma} - 1\right) = E - r_1\left(1 - \frac{x_1^*}{K_1}\right), \tag{5.1}$$

which simplifies to:

$$0 = E - r_1\left(1 - \frac{x_1^*}{K_1}\right), \tag{5.2}$$

so

$$\frac{x_1^*}{K_1} = 1 - \frac{E}{r_1}. \tag{5.3}$$

Similarly, from the second equation we get:

$$\frac{x_2^*}{K_2} = 1 - \frac{E}{r_2}. \tag{5.4}$$

With $x_1^* = \gamma x_2^*$, we now have the system:

$$\begin{aligned} x_1^* &= K_1\left(1 - \frac{E}{r_1}\right), \\ x_2^* &= K_2\left(1 - \frac{E}{r_2}\right), \\ x_1^* &= \gamma x_2^*. \end{aligned} \tag{5.5}$$

Eliminating $x_1^*, x_2^*$ from (5.9), we derive:

$$\gamma = \frac{K_1\left(1 - \frac{E}{r_1}\right)}{K_2\left(1 - \frac{E}{r_2}\right)}. \tag{5.6}$$

Cross-multiplying and simplifying:

$$\gamma\left(1 - \frac{E}{r_2}\right) = \frac{K_1}{K_2}\left(1 - \frac{E}{r_1}\right). \tag{5.7}$$

Multiply out,

$$\gamma - \gamma \frac{E}{r_2} = \frac{K_1}{K_2} - \frac{K_1}{K_2}\frac{E}{r_1},$$

which leads to,

$$E\left(\frac{\gamma}{r_2} - \frac{K_1}{K_2}\frac{1}{r_1}\right) = \gamma - \frac{K_1}{K_2}.$$

Solving for $E$,

$$E^* = \frac{\gamma - \frac{K_1}{K_2}}{\frac{\gamma}{r_2} - \frac{K_1}{K_2}\frac{1}{r_1}} = \frac{\gamma r_1 r_2 - \frac{K_1}{K_2} r_1 r_2}{\gamma r_1 - \frac{K_1}{K_2} r_2}. \tag{5.8}$$

Multiplying numerator and denominator by $K_2$,

$$E^* = \frac{r_1 r_2(\gamma K_2 - K_1)}{\gamma r_1 K_2 - r_2 K_1}. \tag{5.9}$$

This expression gives the effort level $E^*$ for which a coexistence equilibrium exists. For instance, if the patches are identical ($r_1 = r_2 = r$) and ($K_1 = K_2 = K$), then (8) reduces to $\gamma = \frac{1-\frac{E}{r}}{1-\frac{E}{r}} = 1$. Thus $\gamma$ must equal 1 for a coexistence equilibrium, and (9) yields $E^* = \frac{rr(\gamma K - K)}{r\gamma K - rK}$. If $\gamma = 1$, this gives $E^* = \frac{r^2(K-K)}{rK-rK} = 0/0$, an indeterminate form – meaning that any $E < r$ will allow a coexistence equilibrium in the symmetric identical case. In fact, for identical patches, the equilibrium is simply $x_1^* = x_2^* = K(1 - \frac{E}{r})$ for any $E < r$, which obviously satisfies $x_1^* = x_2^*$ and $\gamma = 1$. This is consistent: identical patches with symmetric dispersal act like a single combined logistic population that has a positive equilibrium for any subcritical harvest $E < r$. For non-identical patches, the existence condition (9) implies that an interior equilibrium exists only if $E$ is not too large. If $E$ computed from (9) is negative or exceeds certain values, it indicates no positive solution. Generally, a coexistence equilibrium will exist if and only if $E$ is below a critical threshold. We can identify this threshold by examining when the equilibrium values $x_1^*, x_2^*$ from (7) become zero or negative. From (7), $x_1^* = K_1(1 - \frac{E}{r_1})$ requires $E < r_1$ to have $x_1^* > 0$. Likewise $x_2^* > 0$ requires $E < r_2$. Thus, a necessary condition for coexistence is $E < \min(r_1, r_2)$. If the fishing effort exceeds the intrinsic growth rate of both patches, then neither patch can sustain a positive stock even in isolation, so together they certainly cannot. In fact, if $E \geq r_1$ and $E \geq r_2$, the trivial equilibrium (extinction) is the only possibility. We will confirm this through stability analysis. We therefore expect a critical fishing effort $E_c$ (some value between $r_1$ and $r_2$) beyond which the interior equilibrium disappears (bifurcates into extinction). If $r_1 \neq r_2$, typically the larger growth rate patch can sustain fishing up to its own threshold. Suppose without loss of generality that $r_1 > r_2$. If $r_2 < E < r_1$, patch 2 would go extinct in isolation, but patch 1 could sustain some population. With migration, patch 1 might continue to supply patch 2 with fish, potentially maintaining a two-patch equilibrium even though patch 2 on its own would be overfished. The condition (9) captures this interplay. (In the Stability section, we will see that for $E$ slightly above $r_2$ but below $r_1$, the equilibrium has $x_2^*$ small but still positive due to migration rescue from patch 1.) In summary, the coexistence equilibrium, if it exists, is unique for given parameters, and it satisfies formulas (7) linking it to $E$ and $\gamma$. Existence requires $0 < E < E_c$ for some critical $E_c$ which we will identify via stability (eigenvalue) analysis next.

# 6 Temperature dependent Assymetry

Let's assume patch 1 is the cool area and the temperature of Patch-2 varies . so,the model becomes,

$$\frac{dx_1}{dt} = r_1 x_1\left(1 - \frac{x_1}{K_1}\right) - Ex_1 + D((\gamma_0 + aT)x_2 - x_1), \tag{6.1}$$

$$\frac{dx_2}{dt} = r_2 x_2\left(1 - \frac{x_2}{K_2}\right) - Ex_2 + D(x_1 - (\gamma_0 + aT)x_2). \tag{6.2}$$

The asymmetry parameter $\gamma$ is not a constant but a function of environmental temperature $T$. We are given $\gamma(T) = \gamma_0 + aT$, where $\gamma_0$ and $a(-\frac{\gamma_0}{T} < 0 < \frac{\gamma_0}{T})$ are constants. This formulation means that environmental changes (e.g. ocean temperature changes) can alter migration patterns. The sign of $a$ dictates the direction of change: if $a > 0$, higher temperature $T$ increases $\gamma$, which (for $T$ uniform across the region) tends to increase the relative migration flow from patch 2 to patch 1. If $a < 0$, higher $T$ decreases $\gamma$, favoring movement from patch 1 to patch 2. We assume $T$ is a parameter (a slowly varying environmental variable or a scenario parameter) and not a dynamic variable in the model. Thus, we can analyze the system for different fixed values of $T$ to understand how a climate scenario affects the fish dynamics. It is useful to interpret $\gamma$ in terms of fish behavior. One could imagine patch 1 as a cool-water refuge and patch 2 as a warm habitat. If rising temperatures cause fish to seek cooler waters, one would expect more fish to migrate from patch 2 to patch 1 as $T$ increases – this corresponds to $a > 0$ (increasing $\gamma$ biases net flow toward patch 1). On the other hand, some species might expand into warmer areas when temperatures rise, which would correspond to $a < 0$ (higher $T$ causing a shift of fish from patch 1 to patch 2). Our model can capture either scenario through the sign of $a$. Empirical studies have shown that temperature changes can indeed affect connectivity: for example, in many marine species, especially during larval dispersal, connectivity tends to decrease with higher temperatures (due to faster larval development and shorter dispersal times) . In adult fish, temperature shifts can alter migration timing and routes – for instance, fish may migrate earlier in warmer years or move to maintain their preferred thermal habitat. Thus, $\gamma(T)$ is a simplified way to encode such temperature-driven migration behavior into the model.

- When $a > 0$: Higher temperatures,i,e $\gamma > 1$ increase migration from patch 2 to patch 1.This can concentrate population in patch 1 as temperatures rise.If patch 1 has higher fishing pressure, this could accelerate depletion
- When $a < 0$: when $\gamma < 1$ Higher temperatures increase migration from patch 1 to patch 2.This can shift population toward patch 2 as temperatures rise.May create climate refugia if patch 2 has lower fishing pressure
- When $\gamma \neq 1$: Baseline asymmetry exists even without temperature effects
- If $\gamma > 1$: Natural tendency for fish to move toward patch 1
- If $\gamma < 1$: Natural tendency for fish to move toward patch 2

# 7 Numerical Simulation

This study has analytically and numerically examined a two-patch logistic fishery model, incorporating temperature-dependent asymmetric migration. Through rigorous mathematical analysis supported by numerical illustrations, we demonstrated how migration asymmetry, driven by environmental temperature changes, fundamentally alters equilibrium population distributions and economic profitability. We identified critical conditions under which spatially coordinated management enhances economic yields significantly beyond isolated management practices. These insights not only enrich theoretical understanding of spatial population dynamics but also provide actionable guidelines for marine resource managers facing the complexities of climate-driven ecological change.

The nullcline analysis depicted in **Figure 1** provides essential insights into the equilibrium dynamics of the two-patch system. The blue and green curves represent the nullclines for Patch1 ($x_1$) and Patch2 ($x_2$), respectively, indicating the sets of points in phase space where the population growth rates become zero for each patch independently.

The intersection point (marked by a red dot) of the two nullclines represents the coexistence equilibrium of the coupled system. Vector fields around this equilibrium clearly indicate its local stability, as evidenced by the directional arrows converging toward this intersection. The dashed purple lines, termed K-lines, represent the carrying capacities of each patch, serving as boundaries for the maximum sustainable populations in the absence of migration.

An important ecological interpretation emerges from this analysis: migration dynamics strongly influence equilibrium positions and stability. Specifically, the curvature of the nullclines and their intersection points illustrate how asymmetrical migration redistributes populations, stabilizing the system around a unique coexistence equilibrium. This highlights that migration acts as a balancing mechanism, mitigating localized disturbances and promoting overall ecological resilience.

In summary, the nullcline analysis reinforces the mathematical and ecological conclusions derived earlier, providing a clear visual confirmation of the analytical results regarding equilibrium stability and the role of migration in spatially structured fisheries.

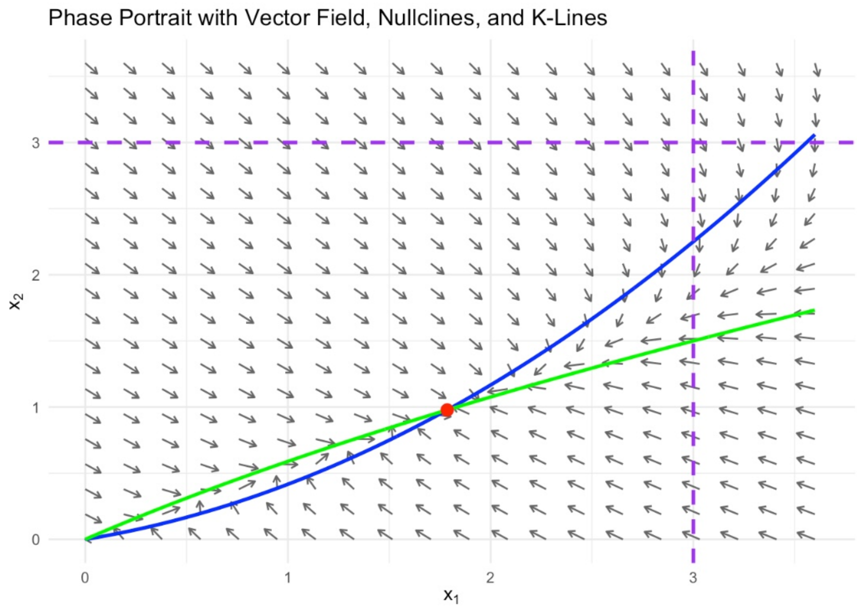


Figure 1: Phase portrait with vector field, nullclines, and carrying capacity (K-lines), illustrating equilibrium stability and the directional influence of migration between patches.

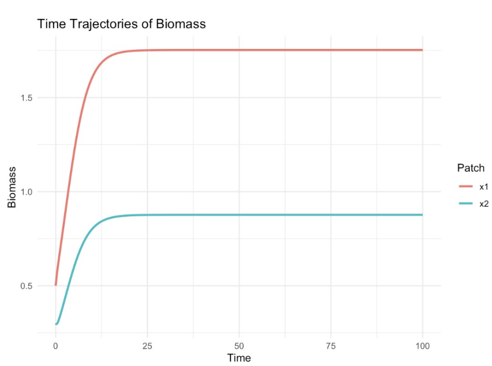


Figure 2: Population vs Time

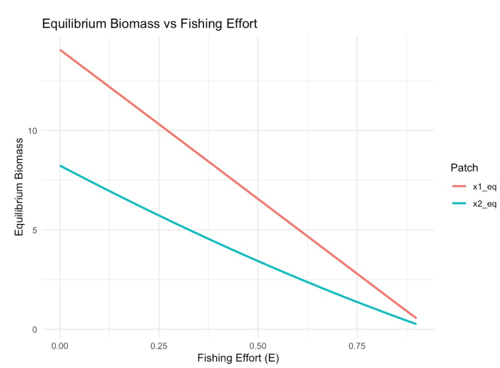


Figure 3: Population vs Fishing effort

Analyzing the population dynamics over time **Figure-2**, reveals distinct transient behaviors and convergence to equilibrium states influenced significantly by migration asymmetry and environmental temperature. Figure-1 demonstrates these dynamics for two interconnected patches at a fixed environmental temperature ($T = 70°\text{C}$). Initially, rapid adjustments in populations of both patches are observable, eventually reaching a stable equilibrium. Patch1, initially supporting a higher population, declines significantly while Patch2, starting from a lower initial density, gradually stabilizes at a higher population level. This clearly illustrates the ecological implication of migration asymmetry: sustained directional migration driven by environmental differences significantly influences local population equilibria. Temperature emerges as a pivotal environmental driver in shaping population equilibria across patches, as illustrated in Figure **??**. A noticeable trend is evident; equilibrium populations in both patches strongly depend on temperature variations. At lower

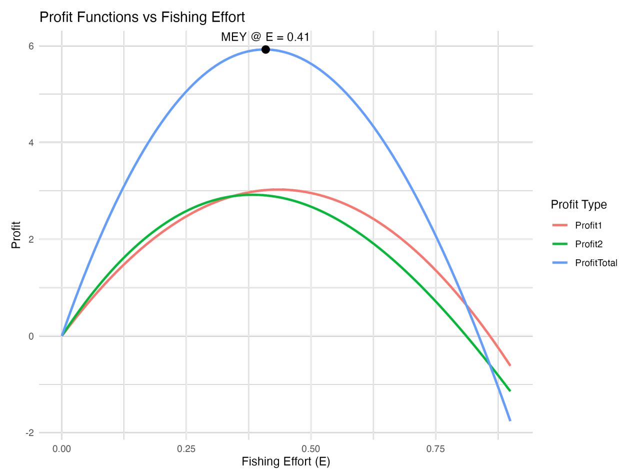


Figure 4: Maximum Economic yeild when the patches are connected

temperatures, Patch1 sustains higher populations compared to Patch2. However, as temperature increases, equilibrium populations in Patch1 decline sharply, whereas those in Patch2 increase, suggesting temperature-dependent migration preferences likely driven by ecological factors, such as resource availability and habitat suitability under changing thermal regimes. This supports ecological expectations that marine species exhibit thermally mediated habitat shifts, often favoring patches providing optimal thermal conditions [10].

The **Figure-2** and **Figure-3** reveals how migration asymmetry (with ($\gamma = 2.0$) and harvesting interact to push population flow, likely favoring Patch 1 over Patch 2. Overall, this phase portrait provides a powerful qualitative tool for understanding the long-term behavior and stability of the system under the given harvesting and migration conditions.

The **Figure-4** illustrates how profits from a two-patch fishery system respond to varying levels of fishing effort (E), showing individual profits from each patch (Patch 1 and Patch 2) and the total combined profit. Initially, as fishing effort increases from low levels, profits rise because more biomass is being harvested without depleting the fish population significantly. However, beyond a certain point, increased effort begins to reduce fish stocks below sustainable levels, leading to diminishing returns and a decline in profits. This trade-off creates a peak in the total profit curve, which represents the Maximum Economic Yield (MEY)—the effort level that maximizes total profit from both patches. The plot clearly identifies this optimal point with a black dot and label. Beyond MEY, continued increases in effort cause overexploitation, lowering the biomass in both patches and thus reducing profits despite higher harvest intensity. The plot highlights the importance of carefully managing fishing effort to balance short-term economic gain with long-term sustainability, and it shows how economic optimization is intrinsically linked to underlying ecological dynamics.

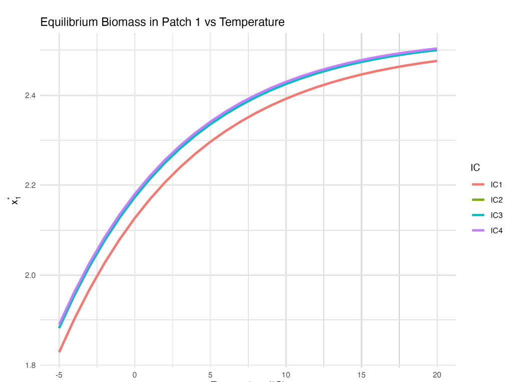


Figure 5: Population,$x_1$ vs Temperature

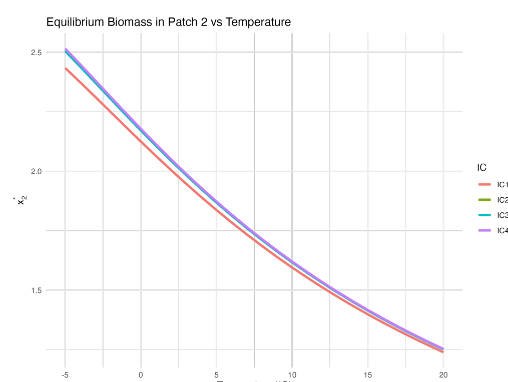


Figure 6: Population,$x_2$ vs Temperature

As temperature increases, the observed trend in the model shows that the population in Patch 1 consistently increases while the population in Patch 2 declines. This pattern arises due to the temperature-dependent migration mechanism modeled through the function $\gamma(T) = \gamma_0 + aT$, where T is temperature. Since $a > 0$, increasing temperature enhances the value of $\gamma$, which biases the movement of individuals more strongly toward Patch 1. Biologically, this can be interpreted as a situation where rising temperatures make Patch 1 increasingly favorable or attractive—perhaps due to better habitat conditions, optimal thermal preferences, or higher resource availability—causing more individuals to migrate into Patch 1 from Patch 2. Conversely, Patch 2 becomes progressively depopulated as temperature rises, not because its intrinsic growth rate declines, but because it loses individuals to Patch 1 at an accelerating rate. This temperature-driven migration skews the spatial distribution of the population, leading to a net accumulation in Patch 1 and depletion in Patch 2. Over time, this can lead to ecological imbalances such as local overpopulation in Patch 1, increased competition, and potential resource depletion, while Patch 2 may face risks of underutilization or even collapse. From a management perspective, this highlights the critical need to account for environmental variables like temperature when assessing population dynamics, spatial harvesting strategies, and conservation planning, especially under climate change scenarios where temperature trends are shifting rapidly.

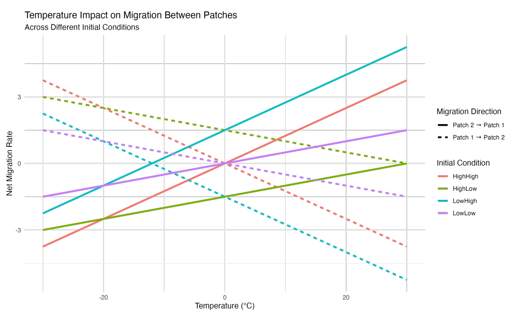


Figure 7: When $\gamma > 1$

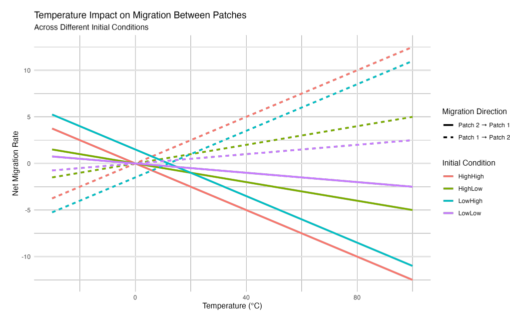


Figure 8: When $\gamma < 1$

The directional migration flows between patches, modulated by the temperature-dependent parameter $\gamma(T)$, are critically examined in **Figures 7** and **Figure-8**.They illustrates how temperature influences net migration rates between two patches under different initial population distributions. As temperature increases, the migration asymmetry parameter $\gamma(T) = \gamma_0 + aT$ decreases due to a negative temperature coefficient ($a < 0$), making Patch 1 progressively less attractive. This shift in preference alters the direction and magnitude of fish movement. In scenarios where Patch 2 starts with more biomass (e.g., LowHigh), there is initially a strong net migration from Patch 2 to Patch 1. However, as temperature rises and $\gamma$ declines, the inflow into Patch 1 weakens and eventually reverses, resulting in fish migrating from Patch 1 to Patch 2. Conversely, when Patch 1 starts more populated (e.g., HighLow), it acts as a source, but the extent of outflow depends on how unattractive it becomes with warming. These plots distinctly indicate the sensitivity of net migration rates to temperature fluctuations. Specifically, Figure 7 highlights that at moderate temperatures (-20–20,°C) and for $\gamma > 1 (a > 0)$, migration from Patch2 to Patch1 intensifies, indicating habitat preference shifts toward Patch1. Conversely, at higher temperatures ($> 30$,°C) when $a < 0$ indicating $\gamma < 1$, this trend sharply reverses, with substantial migration flowing back from Patch1 to Patch2. The reversal points—where net migration changes direction—vary across initial conditions, highlighting the interaction between temperature-driven behavior and initial biomass. Biologically, this shows how climate-driven changes in movement preferences can redistribute populations spatially, with implications for habitat use, local depletion, and management strategies, particularly when patch desirability shifts in response to environmental conditions.

The plot **Figure-9** and **Figure-10** illustrates how total profit from harvesting a two-patch fishery system changes with fishing effort under different temperature scenarios. Each curve corresponds to a different temperature, and shows the profit as effort E increases from zero to near-collapse levels. At low efforts, profits are low because harvesting is minimal; as effort increases, profits rise due to increased harvest, but eventually decline as overfishing reduces biomass. The shape of each curve reflects this classic trade-off. The impact of temperature is introduced through the migration asymmetry parameter $\gamma(T)$, which modifies how

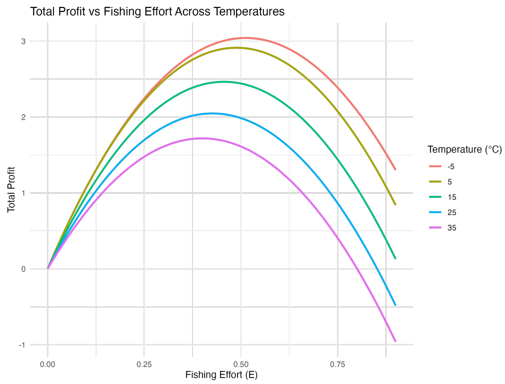


Figure 9: When $E < E_c$

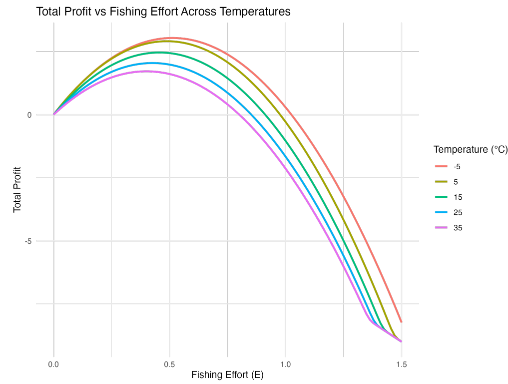


Figure 10: When $E > E_c$

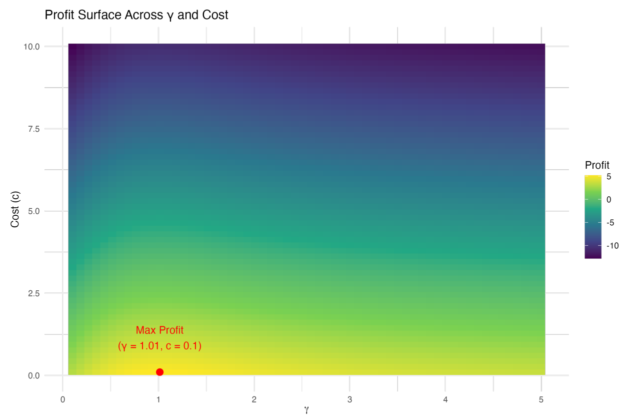


Figure 11: The plot highlights the optimal economic balance achieved through moderate migration asymmetry and manageable harvesting costs and the red dot marks the combination of $\gamma$ and $c$ at which profit is maximized.

fish move between the patches. As the temperature increases, $\gamma(T)$ also increases (since $a > 0$), making Patch 1 more attractive and changing the biomass towards it. This redistribution affects the local stock sizes, and since profits depend on the product of effort and biomass (minus costs), temperature alters both the magnitude and the location of the peak profit (i.e., the optimal effort). In the plot, we observe that higher temperatures tend to shift the profit curves, sometimes lowering peak profit or moving the optimum to lower or higher effort levels. This highlights the importance of accounting for climate-driven migration patterns in economic planning, as ignoring the effects of temperature on fish movement can lead to suboptimal or even unsustainable harvesting strategies.

The plot **Figure-11** illustrates how total profit in a symmetric two-patch fishery system is influenced by changes in the migration asymmetry parameter $\gamma$ and the cost per unit effort $c$. The color gradient represents profit levels across a range of $\gamma$ and $c$, with the red dot marking the point at which profit is maximized. The simulation reveals that profit increases when costs are low and migration asymmetry is moderate, indicating an optimal balance between dispersal efficiency and economic viability. At low values of $\gamma$, migration is too weak to effectively redistribute biomass between patches, leading to under-utilization of productive areas.

# 8 conclusion

This study investigates the dynamics of a spatially structured fishery system composed of two interconnected patches, with the objective of maximizing economic returns under harvesting pressure. Building on classical bioeconomic models, we incorporate asymmetric migration and spatial heterogeneity in growth rates and carrying capacities to explore how connectivity influences the Maximum Economic Yield (MEY). Through a series of numerical simulations, we demonstrate that connecting two heterogeneous fishing patches via migration can lead to a total profit at MEY that exceeds the sum of the MEYs obtained from managing each patch independently. This result underscores a key distinction from the Maximum Sustainable Yield (MSY) framework, where prior work (e.g.[4]) has established that the connecting patches does not improve the total yield in MSY. Our findings reinforce the notion that, while connectivity may not improve biological yield, it can substantially improve economic efficiency under optimal effort levels.

Furthermore, we explore the role of key ecological parameters,namely the dispersal rate $D$ and migration asymmetry $\gamma(T) = \gamma_0 + aT$, in shaping MEY outcomes. We find that higher dispersal rates promote spatial rebalancing of biomass and enhance profit by mitigating local over-exploitation. In addition, when migration asymmetry is temperature dependent, rising temperatures shift biomass distributions toward more favorable habitats, thereby altering the profit-effort landscape. These dynamics are particularly relevant in the context of climate change, as they highlight the importance of incorporating environmentally driven movement behavior into fishery models. In general, the results emphasize that spatial heterogeneity, ecological connectivity, and adaptive dispersal are critical to maximizing long-term economic benefits. From a policy perspective, this suggests that integrated spatial management–such as regulating migration corridors or optimizing effort allocation across connected habitats–can lead to more sustainable and economically advantageous exploitation of marine resources. Collectively, these results convey essential insights for ecological management and policy making. Incorporating temperature-dependent migration into fishery models reveals critical mechanisms that underlie population distributions and economic productivity. Resource managers must explicitly consider these temperature-driven changes when setting harvesting policies and marine spatial planning. Failure to incorporate these dynamics could lead to miscalculations of sustainable harvesting rates, inadvertently undermining both ecological sustainability and economic productivity. These findings strongly advocate spatially and environmentally informed fisheries management practices, adapting dynamically to changing climate regimes to optimize both ecological integrity and economic outcomes.

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